Quang-Vinh Dang
British University Vietnam
Hung Yen, Vietnam
vinh.dq4@buv.edu.vn

5 July 2026

The Editors-in-Chief
Machine Learning
Springer Nature

Dear Editors,

Please consider the enclosed manuscript, "*VOUCH: Anytime-Valid Certificates for Machine Unlearning via Paired-Canary Hypothesis-Betting*," for publication in *Machine Learning*. In keeping with double-blind review, the manuscript itself carries no author information; I am the sole author, and my identity and affiliation appear only in this letter.

The paper asks a question that regulation has made urgent and that current practice cannot answer: when a provider runs an unlearning algorithm to honour a deletion request, can anyone certify—with controlled error—that forgetting took place? Existing evaluations are fixed-sample tests that are unsound as membership evidence, void their guarantees under the continuous monitoring real audits involve, and lean on retrained reference models that are unaffordable at scale. VOUCH removes all three obstacles at once: *paired ghost canaries*, of which a fair coin admits one twin into training and later into the forget set, make the within-pair score difference symmetric under exact unlearning—an exact, distribution-free null over which betting e-processes deliver an anytime-valid confidence sequence on residual membership advantage, an (ε, α)-forgetting certificate, and a revocation alarm. A central finding is that certificates and alarms must compose across deletion waves in opposite directions—consensus versus product—and that the intuitive alternatives falsely certify up to 99.6% of the time. The framework is validated by a two-thousand-seed calibration study under adversarial optional stopping, certification of four unlearning methods on the TOFU and MUSE-News benchmarks across several architectures, and recoverability probes, at a verifier cost two orders of magnitude below shadow-model evaluation. As stated in the manuscript's data availability statement, all code, per-seed results, and the execution harness are publicly released; the repository reference is withheld from the blinded manuscript and is:

`github.com/vinhqdang/VOUCH-Verifiable-Online-Unlearning-Certification-via-Hypothesis-betting`

This manuscript is original, has not been published previously, and is not under consideration elsewhere. I have no conflicts of interest to declare. I believe the combination of game-theoretic statistics with a verifier-generated null, and the resulting first anytime-valid certification framework for machine unlearning, fits *Machine Learning*'s scope well, and I hope you will find it suitable for review.

As requested, the mandatory Contribution Information Sheet is included as part of this letter, beginning on the following page.

Thank you for your consideration.

Sincerely,

Quang-Vinh Dang
British University Vietnam
Hung Yen, Vietnam
vinh.dq4@buv.edu.vn

Contribution Information Sheet

What is the main claim of the paper? Why is this an important contribution to the machine learning literature?

The main claim is that machine unlearning can be given a statistical certificate that is simultaneously (i) exact and finite-sample rather than asymptotic, (ii) valid at every stopping time an auditor chooses to look, not only at one pre-committed sample size, and (iii) free of any retrained reference model or distributional assumption on the model being audited. We obtain this by a single design decision—planting paired ghost canaries at fine-tuning time, twin records where a fair coin admits one into training and withholds the other entirely—coupled to a single inferential engine, betting e-processes from game-theoretic statistics. The coin makes the null hypothesis "unlearning worked" into an exact, distribution-free statement (twin exchangeability) that the provider generates rather than assumes, and the betting engine turns that null into a certificate whose error guarantee survives continuous monitoring by Ville's inequality.

This matters because verification, not optimisation, is the bottleneck the unlearning literature has been converging on: dozens of papers now show that behavioural evaluations of forgetting can be gamed, that quantisation or light fine-tuning resurfaces "erased" knowledge, and that no unique retrained reference even exists for the multi-stage pipelines used in practice. Regulators such as the GDPR and the EU AI Act presume that erasure can be demonstrated, yet the field has had no procedure whose error rate survives the way audits actually happen—continuously, as deletion requests stream in, with an auditor free to stop whenever the evidence looks conclusive. VOUCH is, to our knowledge, the first framework to cast unlearning certification as sequential equivalence testing and to deliver such a certificate at LLM scale. It further shows, and we regard this as an important secondary contribution, that two design choices which look reasonable in isolation—betting a mixture over multiple membership scores, and composing per-wave certificates across a deletion stream by multiplying their e-values—are provably *unsound*, and we give the sound alternatives (per-score consensus; opposite-direction composition for certificates versus alarms) together with the empirical size of the failure they prevent (up to 99.6% false certification for the naive mixture).

What is the evidence you provide to support your claim? Be precise.

The paper proves four theorems and two propositions and tests every one of them empirically, rather than resting the claim on either alone.

Theoretical evidence. We prove (i) that under exact unlearning the paired-canary score difference is an exact fair coin regardless of model, score function, or data distribution (the exact null); (ii) that certifying with the per-score minimum e-process controls false certification at α under a composite null where some scores may still leak, whereas the adaptive-mixture certificate that seems natural does not; (iii) a matching information-theoretic lower bound showing the betting engine is within a constant factor of optimal sample complexity; and (iv) that composing across a stream of deletion waves is only sound in *opposite* directions for the two claims—certificates by all-wave consensus, alarms by e-value product—with a counterexample showing why multiplying certificate e-values across waves is unsound.

Simulation evidence. Over 2,000 seeds under an adversary that peeks after every observation and stops at the first crossing, VOUCH's realised type-I error stays at or below its nominal α, while a fixed-sample binomial test used the same way inflates sevenfold. The same design confirms the composition failure directly: the mixture-sign certificate falsely certifies 99.6% of runs against 0.7% for the per-score minimum, and multiplying per-wave certificate e-values falsely certifies 96% of ten-wave histories against 0% for all-wave consensus. Median certification time tracks the information-theoretic bound within 15%.

End-to-end evidence on real models. We fine-tune, unlearn (gradient ascent, GradDiff, negative preference optimisation), and certify real language models—GPT-2, Pythia-160M, and Phi-1.5, extended to a six-model zoo spanning 0.9M to 5.1B parameters and training years 2019–2026—on the field's own TOFU and MUSE-News benchmarks, three seeds each. Detection of un-unlearned models succeeds in 16 of 18 benchmark runs under the plain sign arm and 18 of 18 under the magnitude-aware arm we introduce, with zero false certificates issued anywhere; genuine unlearning is correctly certified in 17 of 18 runs, the one miss resolved by our own sample-size guidance. Pairing every certificate with a held-out utility reading exposes a failure invisible to forget-only metrics: gradient ascent earns a certificate while held-out loss balloons to 58–286 nats against retraining's 1.5–3.4—it "forgets" by destroying the model. Relearning and jailbreak recoverability probes further revoke two certificates a one-shot evaluation would have passed. A worked example reproduces one real canary pair verbatim, showing what each model actually answers and the score gap the certificate consumes.

**What papers by other authors make the most closely related contributions, and how is your

paper related to them?

Three literatures border this paper, and the closest neighbours in each are treated explicitly in Section 2 of the manuscript. In statistical verification of unlearning, Zhang et al. (2025) establish that membership-inference scores cannot prove training membership unless the inclusion of the data was randomised—the soundness principle our fair-coin canary design is built to satisfy—while Thudi et al. (2022) argue that auditable definitions are a prerequisite for any unlearning guarantee and Zhang et al. (2024) show a dishonest provider can defeat existing verification strategies; Yu et al. (2025) prove that path-dependence makes retrain-equivalence unattainable for multi-stage pipelines, which is why VOUCH's null is internally generated rather than referenced to a retrained model. The nearest direct competitor is Pandey et al. (2025), which gives a hypothesis-testing notion of certified unlearning, but theirs is a fixed-sample test tied to a Gaussian high-dimensional regression regime rather than an anytime-valid, distribution-free procedure at LLM behavioural scale; certified removal (Guo et al., 2020; Sekhari et al., 2021) offers rigorous $(\varepsilon_u, \delta_u)$ bounds but only in convex or near-convex regimes that become vacuous at the scale we target, and we use it as a semantic anchor (any certified algorithm provably passes VOUCH) rather than as a competing method. Cryptographic proof-of-unlearning (Eisenhofer et al., 2025) attests that the correct computation ran, orthogonal to whether forgetting holds statistically. The inferential engine is borrowed from safe, anytime-valid inference—e-values, test martingales, and betting confidence sequences (Ramdas et al., 2023; Grünwald et al., 2024; Waudby-Smith and Ramdas, 2024; Howard et al., 2021)—whose one prior application adjacent to ours is differential-privacy auditing (Steinke et al., 2023; González et al., 2025); that literature lower-bounds the privacy loss of a training algorithm by mounting a successful attack, whereas VOUCH upper-bounds residual influence *after* unlearning by showing no attack in a declared class succeeds—an equivalence test rather than a significance test. Finally, the unlearning-algorithm and benchmark literature we certify against—negative preference optimisation (Zhang et al., 2024), the TOFU (Maini et al., 2024) and MUSE-News (Shi et al., 2025) benchmarks, and the adversarial-recovery findings of Łucki et al. (2025), Zhang et al. (2025), and Deeb and Roger (2024)—is treated throughout as the set of subjects VOUCH certifies rather than as a competing verification method.

Have you published parts of your paper before, for instance in a conference? If so, give details of your previous paper(s) and a precise statement detailing how your paper provides a significant contribution beyond the previous paper(s).

No part of this paper has been published previously in any venue, conference proceedings, workshop, or preprint server. This submission is the first disclosure of the VOUCH framework, its theoretical results, and the accompanying experimental study.